

Moral Tensors and DecisionProofs

Compiling Language into an Auditable, Grounded Safety Layer

Andrew H. Bond

San José State University · IEEE Senior Member

ORCID 0009-0003-2599-6158

Structure-preserving representation before decision contraction.

PyTorch Conference North America 2026 — Responsible AI poster

pip install erisml-compiler · github.com/ahb-sjsu/erisml-compiler · erisml-lib · agi-hpc

1

Don't collapse to a scalar.

harm

rights

consent

privacy

care

Margaret

Daughter

EMS crew

Agency

who is affected · what is owed

who consented · who bears imposed risk

✓ keep the structure

contract to a decision only at the end, and log the contraction: weights · who lost · what residue remains

the moral tensor (who x what is owed)

★ 0.42

one scalar — structure lost

2

pip install erisml-compiler — structure before the scalar

text

segment

extract

canonicalize

MoralGraph

tensorize

DEME

DecisionProof

MoralGraph nodes: stakeholder · act · maxim · commitment · fact · norm — carries a canonical SHA-256 hash

extractor tiers: rule (deterministic, silicon-castable) · LLM (NRP / vLLM + critic) · probe (LaBSE head)

alpha v0.9.0 · 477 tests · MIT · Zenodo DOI 10.5281/zenodo.20659432

3

Four lenses. No silent winner.

one MoralGraph

Consequentialist harm/care tensor
Gini · worst-off · Shapley

Deontic (Kantian) gates · universalizability via Z3 SMT

Virtue Aristotelian habit-consistency

Care ethics Gilligan · Noddings · Tronto

verdicts disagree? → cross_projection_disagreement

defers to a human

"We run four and show you where they conflict — we don't pick one and hide it."

4

Is this 'good'? → Does it preserve the bonds?

Obligation

Claim

Liberty

No-claim

s = correlative swap (agent ↔ patient)

r^2 = deontic negation

s and r^2 commute → V4 (Klein four-group, order 4) — measured

full D4 (order 8) posited — quarter-turns not yet observed

0.0 perfect

0.155 baseline

0.25 warn

0.30 block

Bond Index: does the judgment survive the swap?

multi-rank tensors: party x time x action x coalition

V4 claim machine-checked: Lean 4 + Mathlib (formal/HohfeldV4.lean)

5

Safety in the loop. Fails safe.

Agent planner

Reflex < 100 μs · hard stops

Tactical ErisML · 10-100 ms

Strategic policy + human oversight

ALLOW

REVISE

BLOCK

proof

proof

proof

proof

SHA-256 DecisionProof chain
previous_proof_hash → proof_hash

ethics service times out → rule-based fallback. Never fails open.

between the agent's planner and its actuators

6

What the model says vs what it exhibits

transformer (Qwen2.5-7B, LLaMA, Mistral)

text lens (what it says)

activation lens (what it exhibits)

delta lens compare

requires human review (5 failure modes)

Activation lens: first calibration numbers (09/26). Consumer-metric gating catches 19-21 pp more real behavior changes than raw L2 in matched flag rates. Shaded area: result; replication running. Probe mid-layers: the final-layer β map falls held-out validation.

turboquant-pro (PyPI): PyTorch-native compression HF 1-liner; TurboQuantic-cache! · Triton + Volta kernels · vLLM — 5x KV live here: 3-bit codec on NATS · claims C1-gated · measured false-clear diag (v2.0.0a3)

7

Auditable means you can replay the judgment, and the regime that made it

full_response

NORMAL

EMERGENCY

Margaret

Daughter

EMS crew

Agency

0.185 forbid

0.705 prefer

0.402 neutral

0.902 prefer

0.318 forbid

0.902 prefer

0.200 forbid

0.802 prefer

decision: stand_by (0.834) → full_response (0.804); stand_by inverts to 0.307

elevation gate: detector 0.97 ± 0.90 → GRANTED scope: paramedics only. Auto-reverts. Both edges audited.

conf 0.89 → REFUSED. Fallback: human review plus the EMS call, with no record share. A spurious emergency unlocks nothing.

worst-off shifts Daughter → Margaret (scoped privacy cost) Shapley authority: Margaret-led → EMS-led

Three DecisionProofs: normal, emergency, and the elevation edge

Context elevation is privilege escalation. The elevation is authenticated, bounded, least-privilege, and audited. examples/care_robot_regimes reproduces every number.

One command. Real numbers. A hash you can verify.

\$ python -m erisml.examples.care_robot_regimes

Same action, two regimes: forbid → obligated. Exact Shapley. Authenticated elevation, audited on both edges. Replay the judgment.

1. Structure before contraction — keep the tensor, log the collapse + the residue.
2. Pluralism is the responsible move — four lenses, defer to a human on conflict.
3. Auditable by construction; fails safe, never open.

alpha v0.9.0 · text path solid · activation lens early; first calibration 09/26 (consumer ranks 1-3 real, higher partial · V4 measured, D4 posited (Lean + Mathlib verified) embodied = design target · conversational agents = live